PARTICLE MEASURING SYSTEMS®
a spectris company

5475 Airport Boulevard, Boulder, Colorado 80301-2339
303.443.7100 800.238.1801

Certificate of Calibration

Date: Jul-02-2025 17:34:00

Work Order: 24346-2507012
Customer Name: Ferretti Validaciones

Unit Under Test

Instrument:	LasairIII_310C	Serial Number:	124738
Firmware Version:	PCE: B2B53624 1.9	Calibration Due:	Jul-02-2026

Certification

Particle Measuring Systems certifies that the instrument listed above meets or exceeds manufacturing specifications and meets the requirements of ISO 21501-4:2018. It has been calibrated using equipment and/or standards whose accuracies are traceable to the National Institute of Standards and Technology (NIST), or have been derived from acceptable values of natural physical constants, or by the ratio type of self-calibration. Counting Efficiency determined by digital output.

This Instrument was calibrated in accordance with the Particle Measuring Systems quality system documentation which meets ISO 9001:2015 requirements. The results contained within this calibration certificate relate only to the item calibrated. This certificate may not be reproduced, except in full, without written consent from Particle Measuring Systems.

Instrument Condition As Found

☑ In-Tolerance ☐ Out-of-Tolerance ☐ Operational Failure

Condition As Left

☑ In-Tolerance

Calibration Standards

Procedure Used:	3467 Rev E	Program Used:	1000013097 Rev Q

Performed by

Technician Name: Favio Wainstein

FAVIO J. WAINSTEIN
Director Tecnico
Dirección de Metrología
HITEC S.R.L.

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Reference Information

Date: Jul-02-2025 17:34:00

Work Order: 24346-2507012
Customer Name: Ferretti Validaciones

Unit Under Test

Instrument:	LasairIII_310C	Serial Number:	124738
Firmware Version:	PCE: B2B53624 1.9	Laser Serial Number:	No Change

Particles Used: As Found

Particle Size	Std Deviation	Lot Number	Expiration Date	Uncertainty
0.300 micron	0.007	276145	Jan-31-2027	0.005
0.496 micron	0.009	274678	Dec-31-2026	0.008
0.994 micron	0.010	275827	Jan-31-2027	0.021
4.973 micron	0.050	265485	May-31-2026	0.054

Particles Used: As Left

Particle Size	Std Deviation	Lot Number	Expiration Date	Uncertainty
0.300 micron	0.007	276145	Jan-31-2027	0.005
0.496 micron	0.009	274678	Dec-31-2026	0.008
0.994 micron	0.010	275827	Jan-31-2027	0.021
4.973 micron	0.050	265485	May-31-2026	0.054

References Used

Type	Serial Number	Calibration Due
LasairII-110	65948	Oct-31-2025
Flow Meter	40401638044	Jul-31-2025

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Verification Data Record

Date: Jul-02-2025 17:00:53

Work Order: 24346-2507012
Customer Name: Ferretti Validaciones

Unit Under Test

Instrument:	LasairIII_310C	Serial Number:	124738
Firmware Version:	PCE: B2B53624 1.9	Laser Serial Number:	No Change

Environmental

Temperature:	23 Deg C	Relative Humidity:	21%

Threshold Information

Channel Size (micron)	Measured Threshold/Gain	Received Threshold/Gain	Measurement Type
Ch 1: 0.30	00725 / 0	00726 / 0	Distribution
Ch 2: 0.50	02718 / 0	02744 / 0	Distribution
Ch 3: 1.00	00732 / 1	00738 / 1	Distribution
Ch 4: 5.00	06295 / 1	06336 / 1	Distribution
Ch 5: 10.00	01440 / 2	01437 / 2	Extrapolate
Ch 6: 25.00	07183 / 2	07163 / 2	Extrapolate

As Found Results

Runtime Parameter	Results	Allowed Range	Expanded Uncertainty	Pass/Fail/ Skipped	
Flow Rate	28.9 L/min	26.9 to 29.7 L/min	n/a	Pass	
Flow Error	2.06%	-5.00 to 5.00%	0.90%	Pass	
Zero Count	70.61 Cnt/m³	<= 176.6 Cnt/m³	n/a	Pass	(*1)
Zero Count 95% UCL	162.40 Cnt/m³	<= 176.6 Cnt/m³	n/a	Pass	(*1)
CE(%) @ 0.30	55.6%	30.0% to 70.0%	0.53%	Pass	
CE(%) @ 0.50	98.9%	90.0% to 110.0%	0.62%	Pass	
Resolution @ 0.30	7.4%	0.0% to 15.0%	0.25%	Pass	
Size Error @ 0.30	0.0%	-10.0% to 10.0%	1.8%	Pass	
Size Error @ 0.50	0.5%	-10.0% to 10.0%	2.0%	Pass	
Size Error @ 1.00	0.6%	-10.0% to 10.0%	2.6%	Pass	
Size Error @ 5.00	0.4%	-10.0% to 10.0%	1.6%	Pass	
Size Error @ 10.00	-0.1%	-10.0% to 10.0%	1.9%	Pass	
Size Error @ 25.00	-0.1%	-10.0% to 10.0%	1.9%	Pass	

Design Parameter	Results	Allowed Range	
Sampling Time	0.84%	0.0% to 1.0%	(*2)
Response Rate	0.003%	0.0 % to 0.5%	(*2)
Maximum particle number concentration	48743777 Cnt/m³	10% coincidence loss	(*2)

Note *1: Calculated false count rate adjusted for 95% UCL
Note *2: Tested and verified during design and development

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Calibration Data Record

Date: Jul-02-2025 17:34:00

Work Order: 24346-2507012
Customer Name: Ferretti Validaciones

Unit Under Test

Instrument:	LasairIII_310C	Serial Number:	124738
Firmware Version:	PCE: B2B53624 1.9	Laser Serial Number:	No Change

Environmental

Temperature:	22 Deg C	Relative Humidity:	22%

Threshold Information

Channel Size (micron)	Measured Threshold/Gain	Measurement Type
Ch 1: 0.30	00736 / 0	Distribution
Ch 2: 0.50	02742 / 0	Distribution
Ch 3: 1.00	00741 / 1	Distribution
Ch 4: 5.00	06318 / 1	Distribution
Ch 5: 10.00	01476 / 2	Extrapolate
Ch 6: 25.00	07392 / 2	Extrapolate

As Left Results

Runtime Parameter	Results	Allowed Range	Expanded Uncertainty	Pass/Fail	
Flow Rate	28.2 L/min	27.5 to 29.2 L/min	n/a	Pass	
Flow Error	-0.41%	-3.00 to 3.00%	0.90%	Pass	
Zero Count	0.00 Cnt/m³	<= 7.1 Cnt/m³	n/a	Pass	(*1)
Zero Count 95% UCL	7.06 Cnt/m³	<= 7.1 Cnt/m³	n/a	Pass	(*1)
CE(%) @ 0.30	54.3%	30.0% to 70.0%	0.48%	Pass	
CE(%) @ 0.50	98.8%	90.0% to 110.0%	0.59%	Pass	
Resolution @ 0.30	6.7%	0.0% to 15.0%	0.62%	Pass	
Size Error @ 0.30	-0.26%	-10.0% to 10.0%	1.7%	Pass	
Size Error @ 0.50	-0.62%	-10.0% to 10.0%	2.0%	Pass	
Size Error @ 1.00	-0.75%	-10.0% to 10.0%	2.6%	Pass	
Size Error @ 5.00	-0.59%	-10.0% to 10.0%	1.6%	Pass	
Size Error @ 10.00	-0.79%	-10.0% to 10.0%	n/a	Pass	
Size Error @ 25.00	-0.79%	-10.0% to 10.0%	n/a	Pass	

Design Parameter	Results	Allowed Range	
Sampling Time	0.84%	0.0% to 1.0%	(*2)
Response Rate	0.003%	0.0 % to 0.5%	(*2)
Maximum particle number concentration	48743777 Cnt/m³	10% coincidence loss	(*2)

Note *1: Calculated false count rate adjusted for 95% UCL
Note *2: Tested and verified during design and development

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